WASP-12b

R-0418

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_221227` · created 2026-07-19T07:46:10+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459939.88135 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1141 +/- 0.0071
Transit depth (Rp/Rs)^2	1.3 +/- 0.16 [%]
Orbital Inclination (inc)	82.21 +/- 1.85
Airmass coefficient 1 (a1)	1.1589 +/- 0.0071
Airmass coefficient 2 (a2)	-0.1151 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	None
Optimal Aperture	4.89
Optimal Annulus	8.23
Transit Duration (day)	0.1213 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1141 ± 0.0071 vs 0.1178 (deviation 0.44σ)
Duration fit vs published	0.1213 d vs 0.125 d (deviation 0.7σ)
Mid-time O–C	-1569.9 min
Depth SNR	13.4
Residual scatter	0.61 %

Light curve

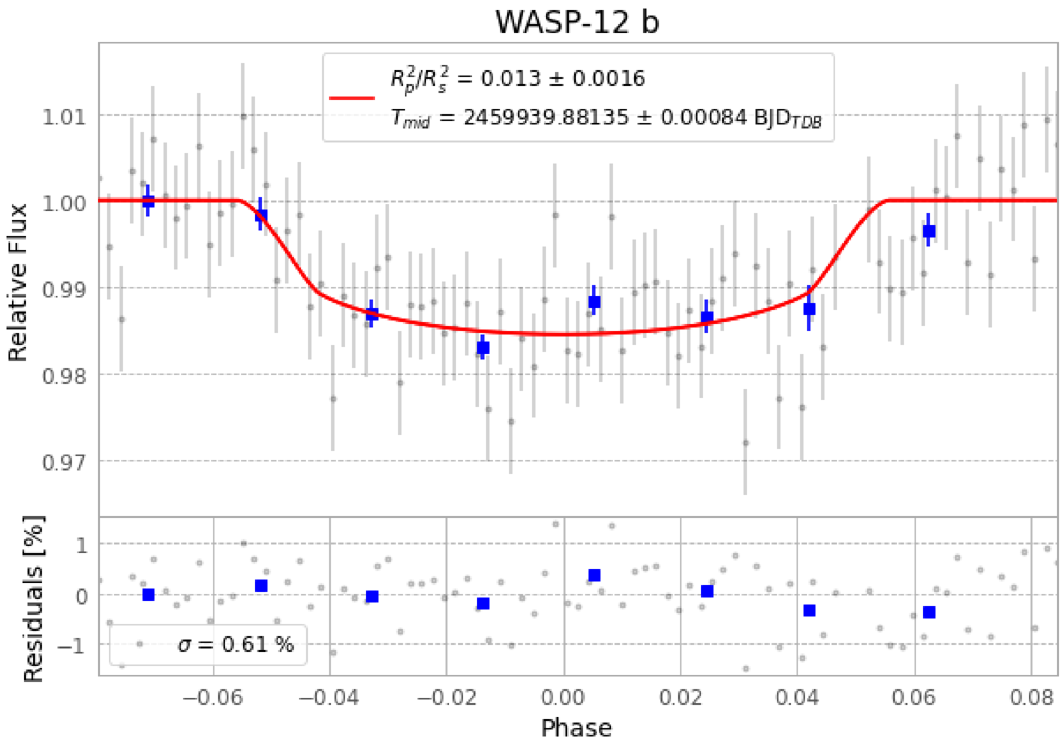

Figure 1 — FinalLightCurve_WASP-12 b_2022-12-25.png (SHA-256 79318ffb98c171c0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 225], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 350c037c320907b3…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

87 FITS frames (57 MB), `WASP-12221226065415.FITS` → `WASP-12221226111215.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221227.log (43bb071aa2be…), FinalLightCurve_WASP-12 b_2022-12-25.png (79318ffb98c1…), FinalLightCurve_WASP-12 b_2022-12-25.csv (cffe025aaf07…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)